



Sebastian Gaviria Trujillo

FULL STACK AI ENGINEER
PYTHON · REACT · GENAI
ML · AGENTIC AI

CONTACT

- sebasga_79@yahoo.com
- +57 310 445 6484
- Sabaneta, Colombia

CORE STACK

BACKEND

- Python
- FastAPI
- Node.js
- Express
- PostgreSQL
- REST APIs

FRONTEND

- React
- Next.js
- TypeScript
- Vite
- HTML5/CSS3

AI / ML

- Generative AI
- Agentic AI
- ML
- LiteLLM
- LLMs
- Scikit-learn
- Pandas

CLOUD / DEVOPS

- AWS
- Azure
- GCP
- Databricks
- Azure DevOps
- Docker
- CI/CD

LANGUAGES

- Spanish Native
- English C1 Certified

TESTING & TOOLS

- pytest
- Playwright
- Jest
- LiteLLM tests
- Load Testing
- Prompt Eng.
- RAG
- SQLAlchemy

Sebastian Gaviria Trujillo

Full Stack AI Engineer | Python · React · Generative AI · ML · Agentic AI

5+ YEARS AI DEV

PRODUCTION LLMS

CLOUD DEPLOYMENTS

BILINGUAL EN/ES

ML & AGENTIC AI

PROFILE SUMMARY

Senior AI Software Engineer with a background in Business Engineering and specialization in Artificial Intelligence. Experienced in designing, developing, and optimizing enterprise solutions based on Generative AI, Machine Learning, and Big Data. Specialized in intelligent agents, RAG systems, LLM orchestration, process automation, and scalable solution deployment on Azure, AWS, GCP, and Databricks. Proficient in Python, FastAPI, React/Next.js, Azure DevOps, MLflow, Unity Catalog, Vector Search, Data Lakes, Data Warehouses, and Lakehouse architectures. Hybrid profile bridging software engineering, data science, and strategic business vision, with the ability to build end-to-end solutions focused on operational efficiency, data-driven decision making, and organizational impact. C1-certified English.

TECHNICAL SKILLS

BACKEND DEVELOPMENT

- Python
- FastAPI
- Node.js
- Express
- REST APIs
- PostgreSQL
- SQLite
- SQLAlchemy

FRONTEND DEVELOPMENT

- React
- Next.js
- TypeScript
- Vite
- HTML5
- CSS3

AI & GENERATIVE AI

- Generative AI
- Agentic AI
- ML
- LiteLLM
- LLM Integration
- Conversational AI
- Prompt Engineering
- RAG & Embeddings
- LangChain
- OpenAI
- Semantic Kernel

DATA & ML

- Machine Learning
- Databricks
- Scikit-learn
- Pandas
- NumPy
- Big Data Pipelines
- Predictive Modeling
- Data Governance

CLOUD & DEVOPS

- AWS
- Google Cloud
- Azure
- Azure DevOps
- Docker
- Kubernetes
- CI/CD
- Cost Optimization

TESTING & QUALITY

- pytest
- Jest
- Playwright (E2E)
- Load & Stress Testing
- API Testing
- Cost Estimation



PROFESSIONAL EXPERIENCE



Senior AI Software Engineer

Apr 2026 – Present

InterScripts · Remote

- ▶ Design, build, and scale enterprise-grade AI systems based on Large Language Models (LLMs) and Generative AI technologies over massive datasets.
- ▶ Implement AI guardrails and responsible AI practices, including security, governance, and access controls in enterprise systems.
- ▶ Develop RAG architectures, model orchestration, semantic retrieval systems, and performance optimization for production AI solutions.
- ▶ Work with frameworks such as LangChain, LlamaIndex, Semantic Kernel, and OpenAI/Azure OpenAI/Anthropic APIs; integrate vector databases (Pinecone, Weaviate, FAISS) and fine-tuning techniques.
- ▶ Collaborate with global teams in a bilingual environment (English/Spanish) across international technical and business teams.
- ▶ Manage MLOps pipelines, AI observability, and scalable deployment patterns with Docker, Kubernetes, and cloud platforms (AWS, Azure, GCP).



Data Scientist (Hourly)

Jan 2023 – Apr 2026

Universidad EAFIT · Medellín, Colombia

- ▶ Developed technology solutions for academic management, operational automation, and advanced analytics, building full-stack apps with Next.js frontends and Python backends (Flask/FastAPI).
- ▶ Integrated Machine Learning models to optimize teacher assignment, availability, and workload; worked with PostgreSQL, SQL Server, and SQLite for internal applications.
- ▶ Designed ETL pipelines for consolidation, transformation, and normalization of academic data; built Streamlit dashboards and analytics tools with Pandas, MLX, scikit-learn, and LLM models.
- ▶ Implemented AI agents for student classification, comment analysis, decision support, and administrative automation, with Azure deployments under scalable architectures.
- ▶ Built quantitative crypto analysis tools connected to Binance API and Bitcoin Core (pruned mode) for order book extraction, liquidity zones, support/resistance, on-chain transactions, volume, and network metrics.
- ▶ Developed predictive models and custom time-series tools based on a personalized Elliott Wave adaptation for crypto markets.



Administrative & Academic Coordinator, Laureles Branch

2024 – Apr 2026

Universidad EAFIT · Medellín, Colombia

- ▶ Drove the business unit to breakeven in 1 year (vs. a 3-year target) and sustained 130% of the revenue goal consistently within 18 months.
- ▶ Deployed AI-powered analytics dashboards for student satisfaction tracking across children, youth, and adult programs using Python, Pandas, and visualization libraries.
- ▶ Built predictive ML models for dropout risk identification, enabling proactive student retention interventions with measurable impact on enrollment KPIs.
- ▶ Engineered Generative AI tools for automated marketing content, academic reporting, and operational analytics using LLM APIs.
- ▶ Led annual budget planning and execution; applied data-driven strategies to optimize resource allocation across academic and administrative functions.



English Teacher · Mentor Teacher · Curriculum Designer

2017 – 2024

Universidad EAFIT · Medellín, Colombia

- ▶ Architected and deployed full-stack AI applications: Generative AI CRM system (FastAPI + PostgreSQL + React), conversational educational assessment chatbot (Next.js + LLM APIs), and an AI Hub for English teachers.
- ▶ Conducted production load and stress testing of LLM endpoints using LiteLLM; optimized inference costs on AWS and GCP deployments with detailed cost estimation reports.
- ▶ Built ML models in Python (Scikit-learn, Pandas) for automated student academic tracking and performance prediction; deployed with FastAPI backend.
- ▶ Developed custom Python software for centralized data management and automatic teacher-to-class assignment, eliminating manual scheduling overhead.
- ▶ Wrote comprehensive pytest test suites for backend services and Jest suites for React frontends; enforced CI/CD pipeline quality gates.
- ▶ Mentored new teachers as senior Mentor Teacher; designed advanced curriculum (B2, Advanced Grammar, Business English).



Co-Founder & General Manager

2014 – 2017

Hiellow SAS · Colombia

- ▶ Led full business lifecycle from ideation to scale: business planning, franchise management (national & international), and investor relations.
- ▶ Directed product innovation and tech development roadmap; managed investment rounds and financial reporting (P&L, balance sheets).
- ▶ Applied early data analysis techniques for market segmentation and performance monitoring.



Academic & Administrative Coordinator

2010 – 2013

Universidad EAFIT · Medellín, Colombia

- ▶ Managed commercial, administrative, and financial operations; prepared annual budgets and KPI reports.
- ▶ Supported ACCET international accreditation process; coordinated teacher selection, training, and development programs.
- ▶ Designed and executed student acquisition and dropout reduction strategies.



Planning & Marketing Coordinator

2004 – 2010

Centro Colombo Americano · Pereira, Colombia

- ▶ Developed institutional strategic plan; managed U.S. Embassy educational projects including financial oversight.
- ▶ Led market research, CRM implementation, financial analysis, and benchmarking initiatives.



Academic & Administrative Coordinator / English Teacher

2001 – 2007

Centro Colombo Americano · Pereira & Cartago, Colombia

- ▶ Taught English across all levels; promoted to coordinator managing a team of 17 teachers, student satisfaction, and marketing for the Cartago branch.



KEY AI PROJECTS & APPLICATIONS

Generative AI CRM System

Full-stack AI-powered CRM for educational client management with LLM-driven insights, automated follow-ups, and analytics dashboards.

FastAPI PostgreSQL React GenAI

LLM Production Load Testing Suite

Comprehensive stress/load testing framework for LLM API endpoints using LiteLLM, with cost calculation reports for AWS and GCP deployments.

LiteLLM pytest AWS GCP

Conversational Assessment Tool

AI-driven conversational app for educational assessment, enabling dynamic question generation and adaptive evaluation for language learners.

Next.js LLM APIs Python Conversational AI

Auto Teacher Assignment Software

Python application for centralized data management and automatic scheduling of teachers to classes, eliminating manual allocation errors.

Python SQLite FastAPI

Student Dropout Prediction Model

ML pipeline for early dropout risk detection using academic and behavioral signals; deployed via FastAPI with automated alerts to coordinators.

Scikit-learn Pandas FastAPI PostgreSQL

AI Hub for English Teachers

AI hub that optimizes English classes through lesson planning, automated AI grading, and content generation for all language skills.

GenAI Agentic AI Lesson Planning LLM APIs



EDUCATION & CERTIFICATIONS



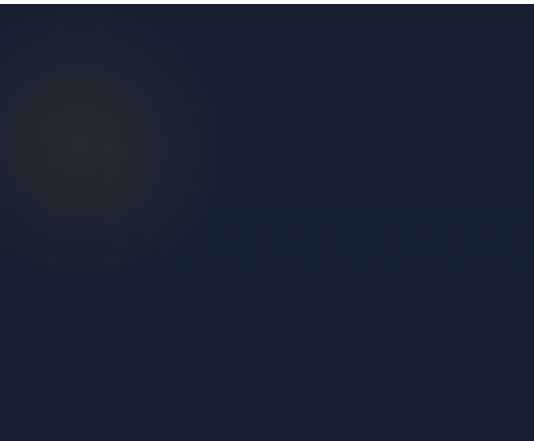
Data Scientist with Python

Platzi · Online
November 2023



Commercial Engineer (License 66149225747RIS)

Universidad Libre · Pereira, Colombia
April 19, 2012



High School Diploma – International Studies

G. Holmes Braddock Senior High School · Miami, Florida, USA

1997



Secondary Education

Colegio De La Salle · Pereira, Colombia

1998



English C1 – Bilingual Certification

Certified English Proficiency · C1 Level

Certified